Cropping Plan and Rotation of a Family Farm

Introduction to Numerical Optimization — Project 2026
Group members: *write the names here*

1 Part A: the coming year

1.1 The linear program

Main assumption. Each hectare carries a single crop during the whole year. The four periods only tell us when the work and the water are needed.

Sets.

- B: the four blocks $B1, \dots, B4$. $\sigma(b)$ is the soil class of block b.
- C: the seven crops (winter wheat, winter barley, grain maize, sugar beet, potato, field beans, cover fallow).
- $T = \{1, 2, 3, 4\}$: the periods P1 spring, P2 early summer, P3 late summer, P4 autumn.
- $C^{\text{sale}} \subset C$: crops that are sold (all except cover fallow).
- C^{min}: crops with a contract minimum (sugar beet).
- C^{max}: crops with a contract ceiling (sugar beet, potato).
- C^{spot}: crops with a free-market price (sugar beet, potato).
- C^{np}: non-productive crops (cover fallow).

Parameters. All of them are read from the nine data files. $A = \sum_b A_b = 80$ ha is the farm area.

Symbol	Meaning	Unit
A_b	area of block b	ha
$y_{c,s}$	yield of crop c on soil class s	t/ha
k_c	variable cost of crop c	EUR/ha
m_c	largest share of the farm that crop c may use	—
ν	smallest share of the farm left non-productive	—
p_c	price of crop c (contract price for sugar beet and potato)	EUR/t
q_c	free-market (spot) price, $c \in C^{\text{spot}}$	EUR/t
$\underline{M}_c$, $\overline{M}_c$	contract minimum and contract ceiling	t
$w_{c,t}$	irrigation needed by crop c in period t	m^3/ha
$l_{c,t}$	labour needed by crop c in period t	h/ha
I_t	inflow into the reservoir in period t	m^3
U_t	pump capacity in period t	m^3
$\bar{R}$, R_0, R_{end}	reservoir maximum, start level, required end level	m^3
L	abstraction permit per year	m^3
π	pumping cost	EUR/m^3
F_t	family hours in period t	h
H_t	largest number of hired hours in period t	h
ω_t	wage of hired labour in period t	EUR/h

Variables. All variables are nonnegative.

- $x_{b,c}$: hectares of crop c on block b (ha).
- s_c: tonnes of crop c sold at the price p_c, for $c \in C^{\text{sale}}$.
- u_t: water pumped in period t (m^3), with $u_t \le U_t$.
- r_t: reservoir level at the end of period t (m^3), with $r_t \le \bar{R}$.
- e_t: inflow lost because the reservoir is full (m^3).
- h_t: hired hours in period t (h), with $h_t \le H_t$.

To keep the formulas short we also use four quantities that are computed from x:

$$X_c = \sum_b x_{b,c}, \qquad Q_c = \sum_b y_{c,\sigma(b)}\, x_{b,c}, \qquad W_t = \sum_c w_{c,t} X_c, \qquad N_t = \sum_c l_{c,t} X_c.$$

They are the area of crop c on the farm, the tonnes produced, the water used for irrigation in period t, and the work hours needed in period t. The tonnes sold on the spot market are $Q_c - s_c$.

Objective. We maximize the gross margin:

$$\max \underbrace{\sum_{c \in C^{\text{sale}}} p_c s_c + \sum_{c \in C^{\text{spot}}} q_c\,(Q_c - s_c)}_{\text{revenue}} - \underbrace{\sum_c k_c X_c}_{\text{crop costs}} - \underbrace{\sum_t \omega_t h_t}_{\text{hired labour}} - \underbrace{\pi \sum_t u_t}_{\text{pumping}}.$$

Constraints.

(C1) **Land.** $\sum_c x_{b,c} = A_b$ for every block b.
Every hectare of a block is used. Land we do not want to farm goes to cover fallow.

(C2) **Diversification.** $X_c \le m_c\, A$ for every crop c.
No crop may cover more than its allowed share of the farm, so the farm does not depend on one crop.

(C3) **Non-productive area.** $\sum_{c \in C^{\text{np}}} X_c \ge \nu A$.
At least 4% of the farm (3.2 ha) is left non-productive.

(C4) **Sales.** $s_c \le Q_c$ for $c \in C^{\text{sale}}$; $s_c \ge \underline{M}_c$ for $c \in C^{\min}$; $s_c \le \overline{M}_c$ for $c \in C^{\max}$.
We cannot sell more than we grow. Sugar beet must deliver its contract minimum. The contract price is paid only up to the ceiling; what is left is sold on the spot market.

(C5) **Reservoir balance.** $r_t = r_{t-1} + I_t + u_t - W_t - e_t$ for $t \in T$, with $r_0 = R_0$.
The water at the end of a period is the water we had, plus the inflow and the pumped water, minus the irrigation, minus the water lost by overflow.

(C6) **Reservoir capacity.** $0 \le r_t \le \bar{R}$.
The water level cannot be negative and cannot be higher than the size of the reservoir.

(C7) **Reserve.** $r_4 \ge R_{\text{end}}$.
The reservoir must be well filled at the end of the year.

(C8) **Pump and permit.** $0 \le u_t \le U_t$ and $\sum_t u_t \le L$.
The pump has a limit per period, and the permit limits the water taken from the stream in the whole year.

(C9) **Labour.** $N_t \le F_t + h_t$ and $0 \le h_t \le H_t$.
The work needed in a period is done by the family hours and by hired hours, and hiring is limited.

Modelling remarks.

- Inside one period, inflow, pumping and irrigation are added together. The capacity applies to the level at the end of the period.
- The variable e_t allows the model to throw away water that does not fit in the reservoir, as the statement says.

- In the data the contract price is higher than the spot price (sugar beet 38 against 22, potato 160 against 92). So the model sells under the contract first, and no extra rule is needed to force this.

1.2 Solution

The model is written in Julia with JuMP and solved with HiGHS. It has 50 variables and 35 constraints (not counting the simple bounds). The solver finds an optimal solution.

Gross margin. The optimal gross margin is **122,077.77 EUR**.

	EUR
Revenue	248,525.10
Variable crop costs	−122,547.32
Pumping cost	−900.00
Hired labour	−3,000.00
Gross margin	122,077.77

Cropping plan block by block. Areas are in hectares. Each row adds up to the area of the block.

Block	Wheat	Barley	Maize	Sugar beet	Potato	Beans	Fallow	Total
B1 (deep loam)	—	—	—	11.11	14.89	—	—	26.00
B2 (silt loam)	18.00	0.85	—	3.15	—	—	—	22.00
B3 (sandy loam)	—	11.30	3.50	—	—	—	3.20	18.00
B4 (heavy clay)	14.00	—	—	—	—	—	—	14.00
Farm	32.00	12.15	3.50	14.26	14.89	0.00	3.20	80.00
Share of farm	40.0%	15.2%	4.4%	17.8%	18.6%	0%	4.0%	100%

Winter wheat is grown on exactly 40% of the farm, its largest allowed share. Potatoes are grown on the deep loam block B1, where their yield is highest, and not on heavy clay, where it is very low. Field beans are not grown.

Production and sales. Everything that is grown is sold. Only the sugar beet and potato contracts have limits.

Crop	Tonnes	Price (EUR/t)	Revenue (EUR)
Winter wheat	302.40	210	63,504.00
Winter barley	98.72	195	19,249.47
Grain maize	42.04	195	8,198.24
Sugar beet	1,199.30	38	45,573.39
Potato	700.00	160	112,000.00

The potato contract ceiling (700 t) is reached exactly. The sugar beet delivery (1,199 t) is above the minimum of 900 t and below the ceiling of 1,700 t, so no spot sales occur.

Use of the reservoir.

Period	Inflow	Pumped	Pump limit	Irrigation	Lost	Level at end
Start						9,000
P1 spring	5,500	0	5,000	0	1,500	13,000
P2 early summer	2,000	4,000	5,000	13,786	0	5,214
P3 late summer	1,200	5,000	5,000	11,414	0	0
P4 autumn	9,000	1,000	5,000	0	0	10,000

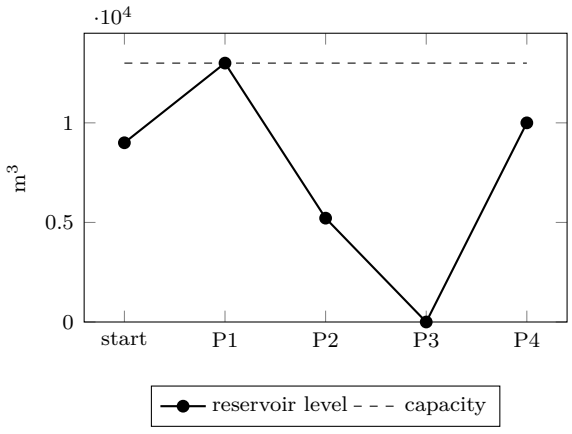
All values are in m^3.

- In spring the reservoir is already almost full (9,000 + 5,500 > 13,000), and no crop needs irrigation, so 1,500 m^3 of free water is lost. Nothing is pumped in spring.
- In the two summer periods irrigation uses 25,200 m^3 in total. It comes from the stock stored in spring, from the inflow, and from 9,000 m^3 pumped. The reservoir is *empty* at the end of P3.
- In autumn the inflow (9,000 m^3) refills the empty reservoir, and 1,000 m^3 more are pumped to reach the required end level of 10,000 m^3.
- The whole permit of 10,000 m^3 is used (pumping cost 900 EUR).

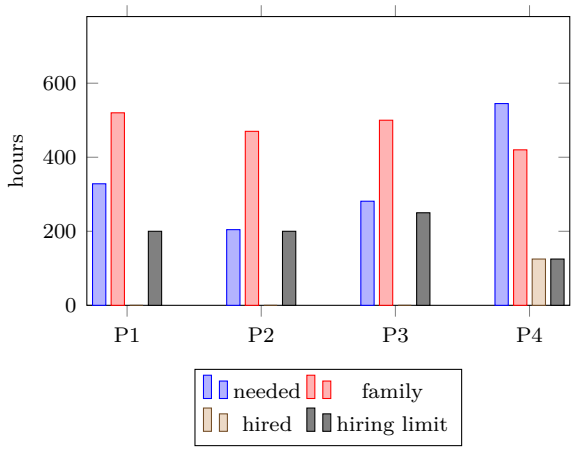
Use of hired labour.

Period	Hours needed	Family hours	Hired	Hiring limit	Wage bill (EUR)
P1 spring	328.3	520	0	200	0
P2 early summer	204.4	470	0	200	0
P3 late summer	281.2	500	0	250	0
P4 autumn	545.0	420	125	125	3,000

The family is enough in the first three periods. Hired labour is used only in autumn, when the potato harvest needs 18 hours per hectare. All 125 hired hours are used (the hiring limit), and they cost 24 EUR per hour.



(a) Reservoir level at the end of each period



(b) Work needed, family hours and hired hours

Figure 1: How the optimal plan uses the reservoir and the hired labour.

Which resources are exhausted?

Exhausted (at the limit)	Not exhausted (slack left)
Land: all 80 ha are sown	Pump in P1 (0 of 5,000), P2 (4,000 of 5,000) and P4 (1,000 of 5,000)
Abstraction permit: 10,000 of 10,000 m^3	Family hours in P1, P2, P3 (slack 192, 266 and 219 h)
Pump in P3: 5,000 of 5,000 m^3	Hired hours in P1, P2, P3 (none needed)
Reservoir full at the end of P1 (13,000 m^3)	Share limit of barley, maize, sugar beet, potato (slack 19.9, 28.5, 17.7 and 17.1 ha)
Reservoir empty at the end of P3	Sugar beet: contract ceiling (1,199 of 1,700 t) and minimum (1,199 of 900 t)
Reservoir end level: exactly 10,000 m^3	Reservoir capacity in P2 and P4
Family hours in P4: 420 of 420 h	
Hired hours in P4: 125 of 125 h	
Share limit of winter wheat: 32 of 32 ha	
Potato contract ceiling: 700 of 700 t	
Non-productive area: 3.2 of 3.2 ha	

In short, water is the main shortage (permit, summer pump and empty reservoir), and autumn labour is the second one. The plan also meets the wheat share limit and the potato contract ceiling.

2 Part B: what the plan is worth

3 Part C: the rotation